Experiment 9

Aim

To develop an Android application using controls like Button, TextView, and EditText for designing a calculator that performs basic operations such as Addition, Subtraction, Multiplication, and Division.

Procedure

1. Open Android Studio and create a new project using **Empty Views Activity** or **Empty Activity**.
2. Design the user interface using:
   - **EditText** for input numbers
   - **Button** for operations (+, −, ×, ÷)
   - **TextView** to display the result
3. Assign unique IDs to all UI components in the XML layout file.
4. Open the main activity file (Java/Kotlin).
5. Link UI components using findViewById() (XML) or state variables (Compose).
6. Write logic for:
   - Addition
   - Subtraction
   - Multiplication
   - Division
7. Handle special cases like:
   - Empty input
   - Division by zero
8. Display the result in the TextView.
9. Run the application using an emulator or real device.
10. Test all operations and verify outputs.

Code

XML Layout (activity_main.xml)

```XML
<?xmlversion="1.0" encoding="utf-8"?>
<LinearLayout

    xmlns:android="http://schemas.android.com/apk/res/android"
    android:orientation="vertical"
    android:padding="20dp"
    android:layout_width="match_parent"
    android:layout_height="match_parent">

    <EditText
        android:id="@+id/num1"
        android:hint="Enter    first    number"
        android:inputType="numberDecimal"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"/>

    <EditText
        android:id="@+id/num2"
        android:hint="Enter second number"
        android:inputType="numberDecimal"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"/>

    <TextView
        android:id="@+id/result"
        android:text="Result"
        android:textSize="24sp"
        android:padding="10dp"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"/>

    <Button
        android:id="@+id/add"
        android:text="Add"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"/>
```

```
    <Button
        android:id="@+id/sub"
        android:text="Subtract"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"/>

    <Button
        android:id="@+id/mul"
        android:text="Multiply"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"/>

    <Button
        android:id="@+id/div"
        android:text="Divide"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"/>

</LinearLayout>
```

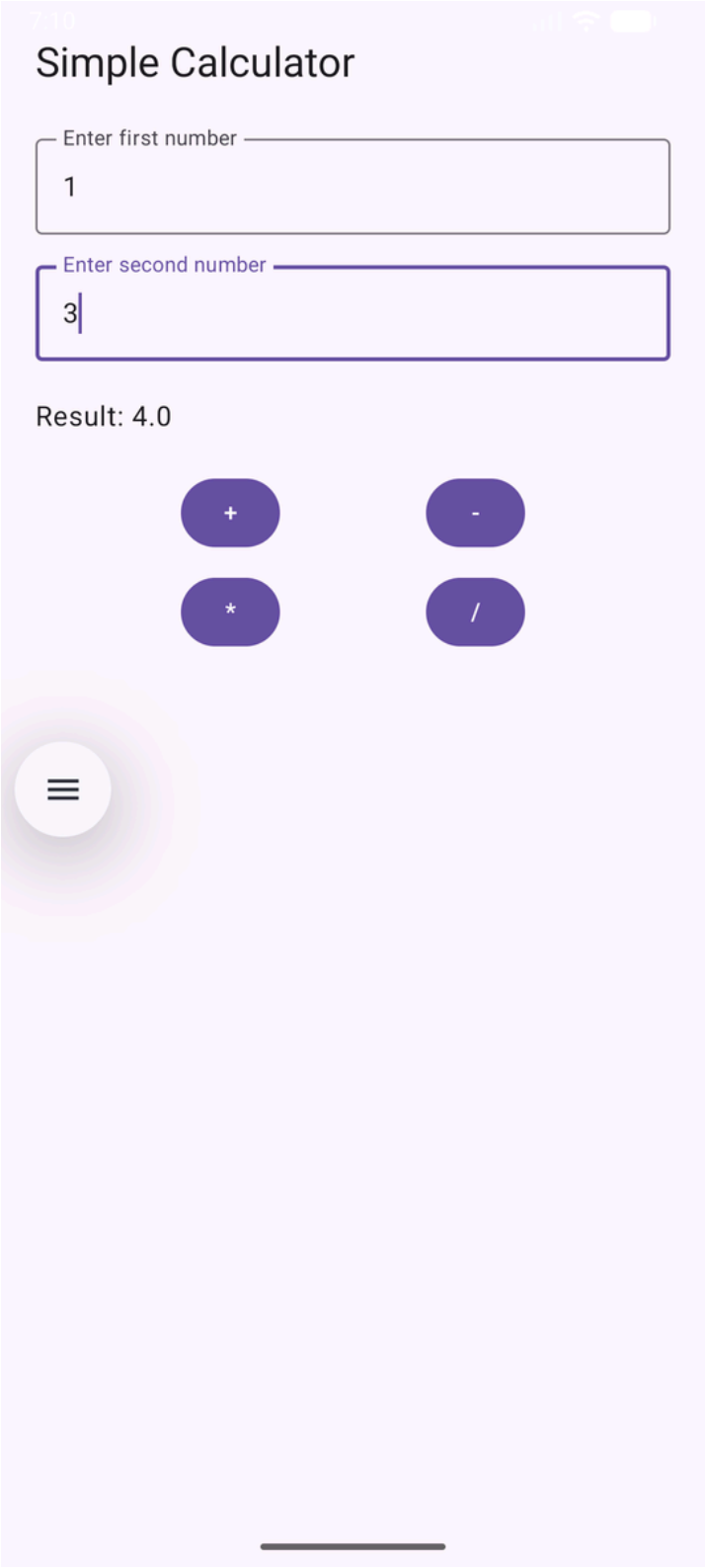
Java Code (MainActivity.java)

```java
package com.example.experiment_9;


import android.os.Bundle;
import android.widget.*;
import androidx.appcompat.app.AppCompatActivity;

public class MainActivity extends AppCompatActivity{

    EditText num1, num2;
    TextView result;
    Button add, sub, mul, div;
```

```
@Override
protectedvoidonCreate(Bundle savedInstanceState){
    super.onCreate(savedInstanceState);
    setContentView(R.layout.activity_main);

    num1=findViewById(R.id.num1);
    num2=findViewById(R.id.num2);
    result=findViewById(R.id.result);

    add=findViewById(R.id.add);
    sub=findViewById(R.id.sub);
    mul=findViewById(R.id.mul);
    div=findViewById(R.id.div);

    add.setOnClickListener(v  ->  calculate("+"));
    sub.setOnClickListener(v  ->  calculate("-"));
    mul.setOnClickListener(v  ->  calculate("*"));
    div.setOnClickListener(v -> calculate("/"));
}

privatevoidcalculate(String op) {
    Stringval1=num1.getText().toString();
    Stringval2=num2.getText().toString();

    if(val1.isEmpty() || val2.isEmpty()) {
    result.setText("Enter values");
    return;
    }

    doublen1=Double.parseDouble(val1);
    doublen2=Double.parseDouble(val2);

    double res;

    switch (op) {
        case"+":res = n1 + n2; break;
```

```
            case"-":res=n1-n2;break;
            case"*":res=n1*n2;break;     case
            "/":
                if (n2 == 0) {
                    result.setText("Cannot divide by zero");
                    return;
                }
                res = n1 / n2;
                break;
            default: res = 0;
        }

        result.setText("Result:"+res);
    }
}
```

Output



Result

Thus, an Android application using Button, TextView, and EditText was successfully developed to perform basic arithmetic operations such as addition, subtraction, multiplication, and division.